Aditya Purohit

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EXPERIENCE

Project Intern – Recruitment Assessment Centre (RAC), DRDO Jun 2025 – Present

- Designed AI/ML-driven recruitment automation systems for candidate shortlisting, expert mapping, and interview scheduling.
- Built semantic matching pipelines using embeddings to align interviewer expertise with candidate skill profiles.
- Automated high-volume recruitment workflows, improving efficiency and scheduling accuracy in defense-grade environments.

Tech Intern – Formskart Jun 2025

- Developed an AI-based college recommendation system using machine learning and Python.
- Built a production-ready Streamlit application recommending colleges based on JEE rank and preferences.
- Implemented MySQL-backed data pipelines and interactive analytics dashboards.

PROJECTS

Pravah – AI-Powered Smart Traffic Management System Smart India Hackathon 2025 – Winner

- Architected a multi-objective PPO reinforcement learning system to dynamically optimize traffic signal phases.
- Achieved **50% improvement in traffic efficiency** and **10.5% CO_2 emission reduction**.
- Built YOLO + OpenCV vision pipelines integrated with SUMO to prioritize emergency vehicles, increasing throughput by **26%**.
- Developed a FastAPI backend with LSTM forecasting to predict congestion 20 minutes ahead, reducing peak congestion by **15–20%**.

MIRA – AI System for Matching Interviewers with Resumes Smart India Hackathon 2024 – Winner

- Built an LLM-based expert panel matching system using Mistral-7B, SBERT, and BERT embeddings.
- Designed automated resume parsing and ETL pipelines, reducing screening time by **40%**.
- Implemented dense vector search for low-latency contextual recommendations at scale.

IVMS – Intelligent Vehicle Monitoring System (ANPR & Vehicle Analytics)

- Engineered a real-time, GPU-accelerated computer vision pipeline sustaining **30+ FPS** inference using PyTorch and OpenCV.
- Built a high-accuracy ANPR system using EAST text detection and TrOCR, achieving **95%+ OCR accuracy**.
- Delivered a production-ready analytics platform with CLIP-based zero-shot color classification and a Streamlit + SQLite dashboard.

EDUCATION

Bennett University, Greater Noida, India Bachelor of Technology (B.Tech) in Computer Science **2023 – 2027**

SKILLS

Programming: Python, C++, Java, JavaScript
AI/ML: Machine Learning, Deep Learning, Reinforcement Learning (PPO), NLP, LLMs, Generative AI, RAG, Agentic AI
Computer Vision: OpenCV, YOLO, CNNs, ANPR, CLIP
Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
Backend/Web: FastAPI, Django, Streamlit, HTML, CSS, JavaScript
Databases: MySQL, MongoDB, SQLite, Vector Databases (FAISS)
Tools/Cloud: Git, GitHub, Jupyter, Google Colab, Anaconda, AWS (fundamentals)

ACHIEVEMENTS & AWARDS

- **International Agentic AI Hackathon 2025** – International Winner (University of Derby, UK)
- **Global CyberAI Hackathon 2025** – Global Winner (Ulster University, UK)
- **Smart India Hackathon 2024 & 2025** – 2× National Winner (Government of India)
- **Student Innovation Excellence Award 2025** – Times Now Education Summit
- **Microsoft AI Innovate 2025** – 1st Runner-up (Top 1.7% of 300+ teams)
- **Smart BUHackathon 2024 & 2025** – Top 1.75% (Ranked 15 & 8 of 400+ teams)